



CLIMATE CHANGE & TICK MIGRATION

Presented By: Ticknicians



OUR TEAM



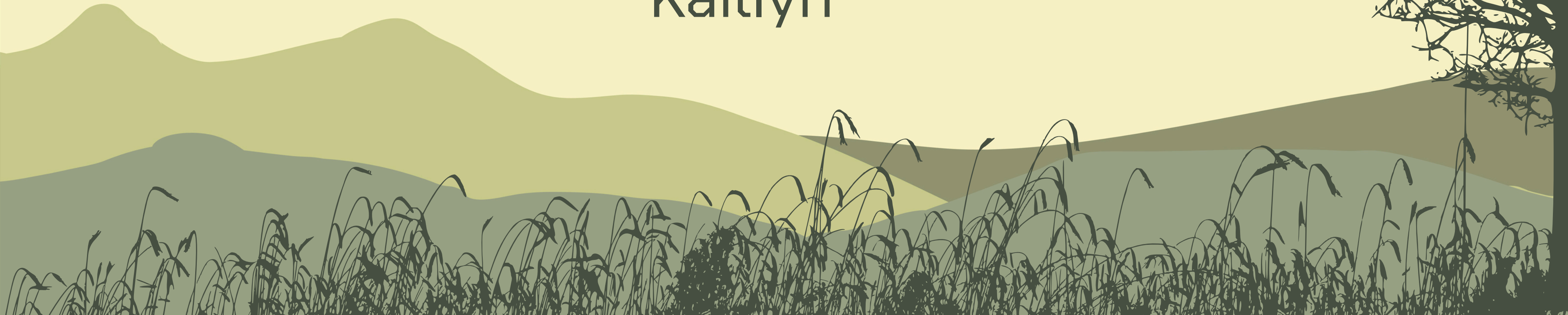
Shaarav

Tyler

Natalie

Shobha

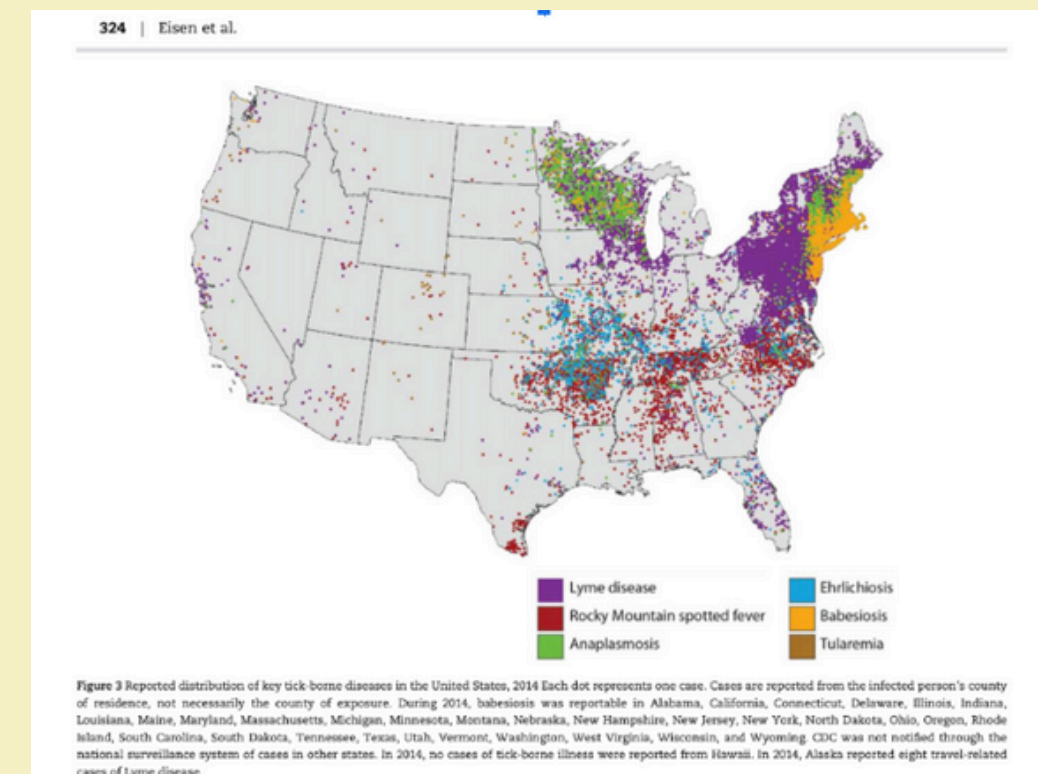
Kaitlyn



PROJECT OVERVIEW

Studies suggest that the range of many tick species is expanding, and that increasing temperature as well as other changes in climate may be influential. Ticks are shown to prefer areas with thick forest, and warm, humid climates. (Soneshine, 2018)

This expansion poses a public health risk because more areas across the U.S. will be exposed to harmful diseases. Ticks spread Lyme Disease, Alpha-Gal syndrome, Rocky Mountain spotted fever, and other diseases to humans.

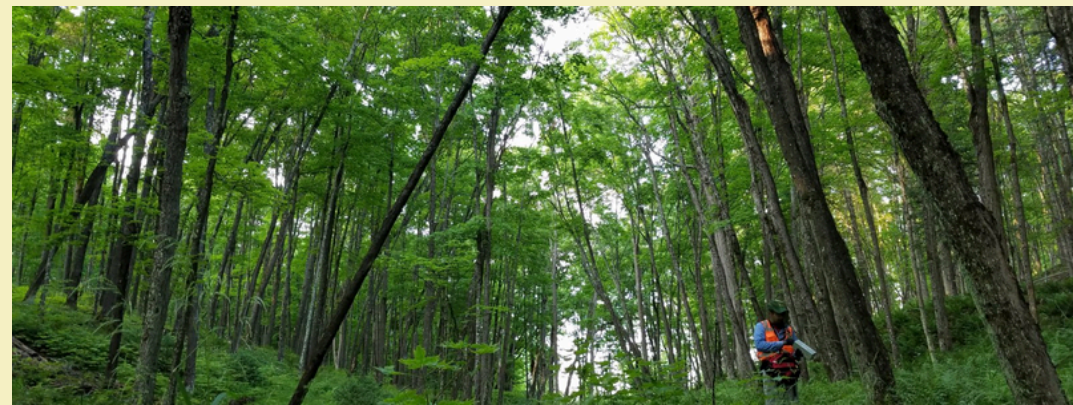


PROJECT OVERVIEW

We chose to look at 3 NEON sites in different regions of the U.S. to compare how their climates may be affecting tick spread.



HARV - Harvard forest, 65 miles west of Boston.
Mean Temp - 45°F



UNDE - University of Notre Dame, Northern Wisconsin.
Mean Temp - 40°F



ORNL - Oak Ridge, Eastern Tennessee.
Mean Temp - 58°F

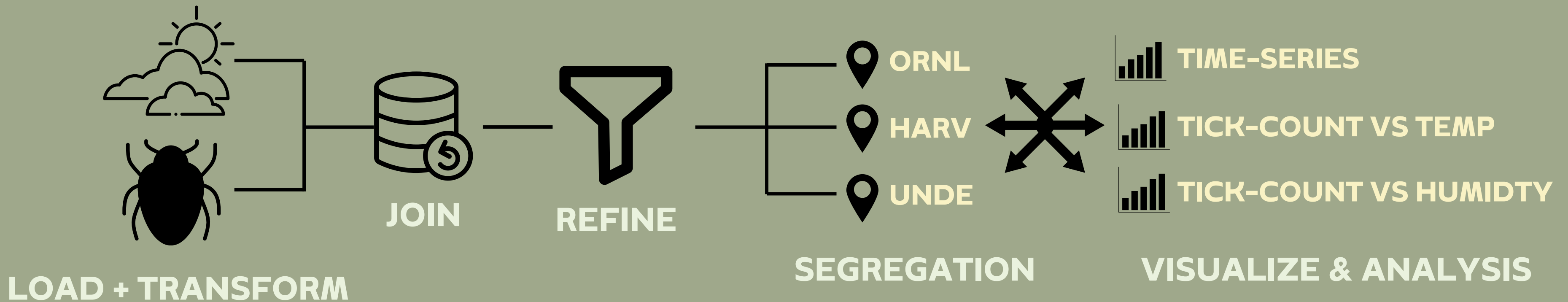




HYPOTHESIS

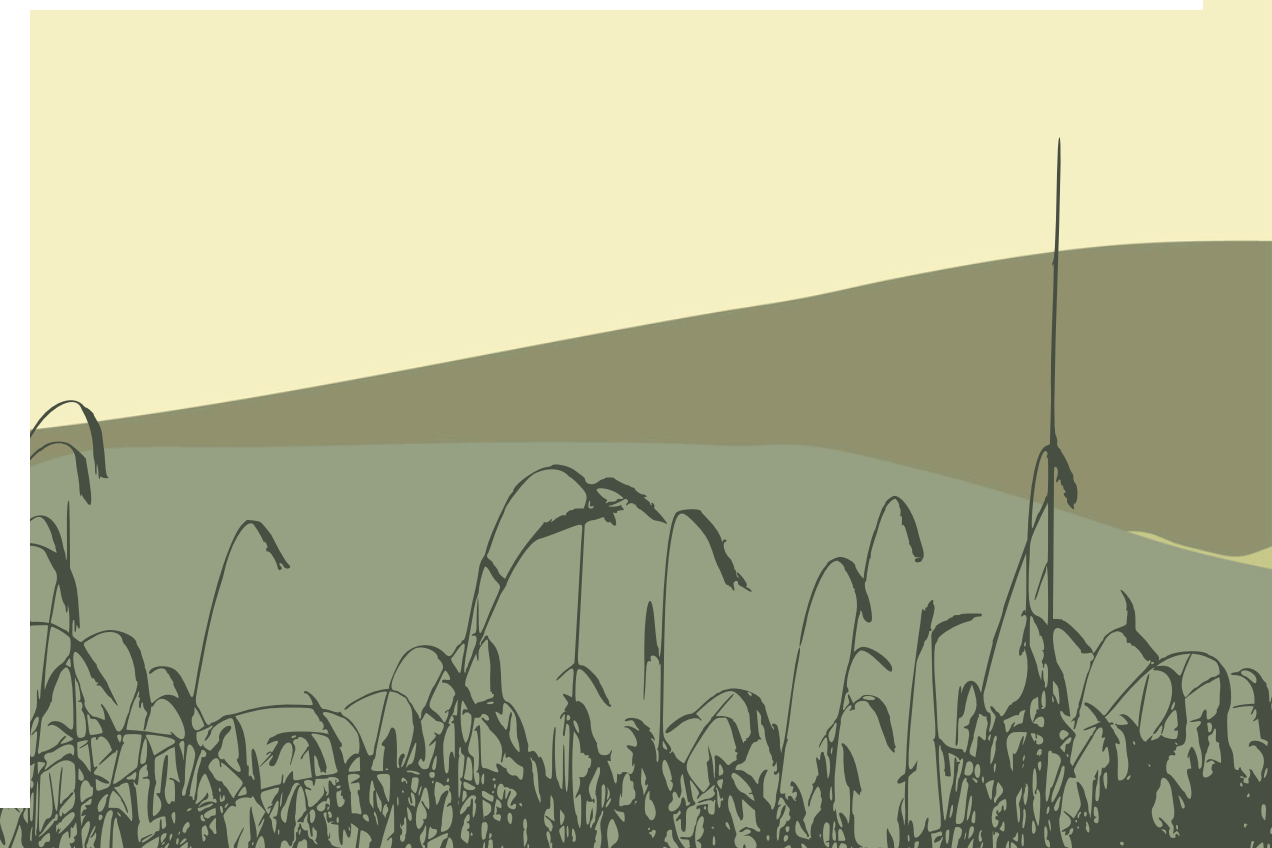
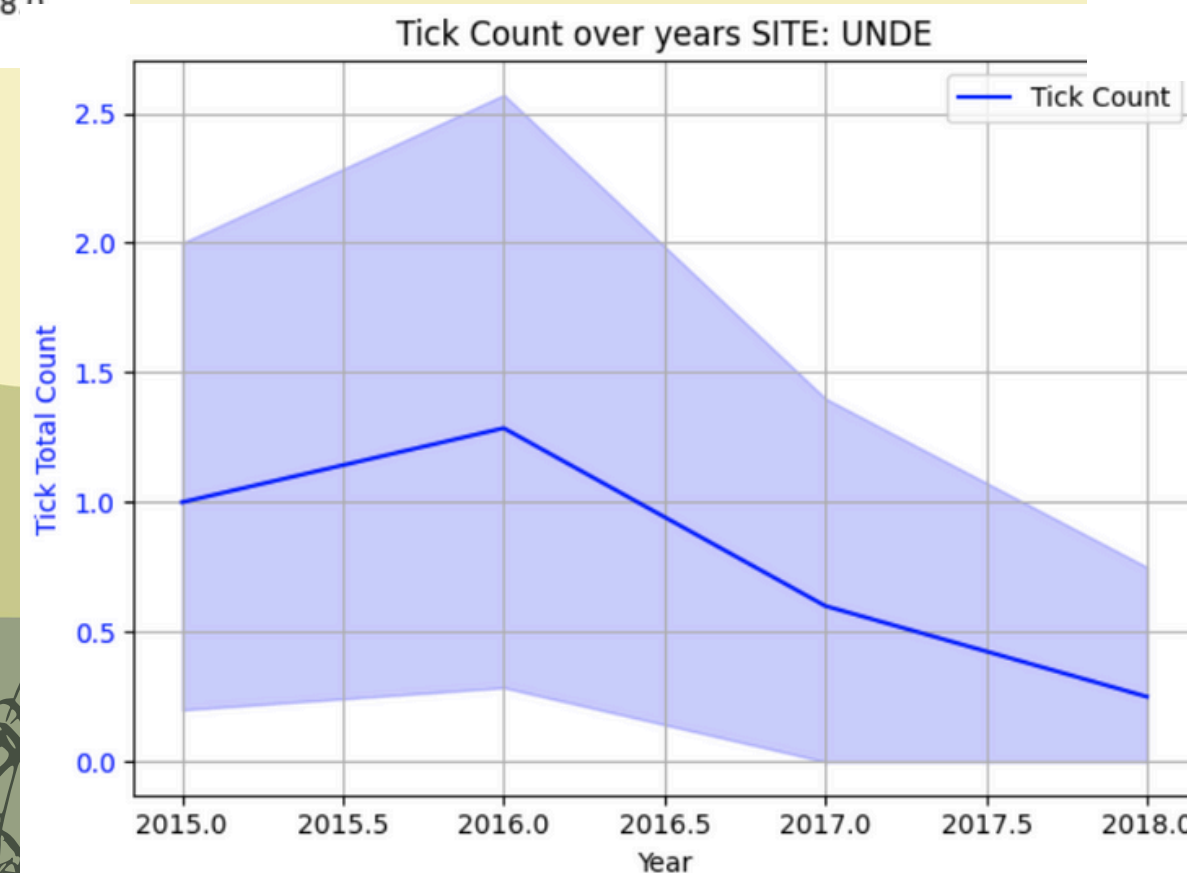
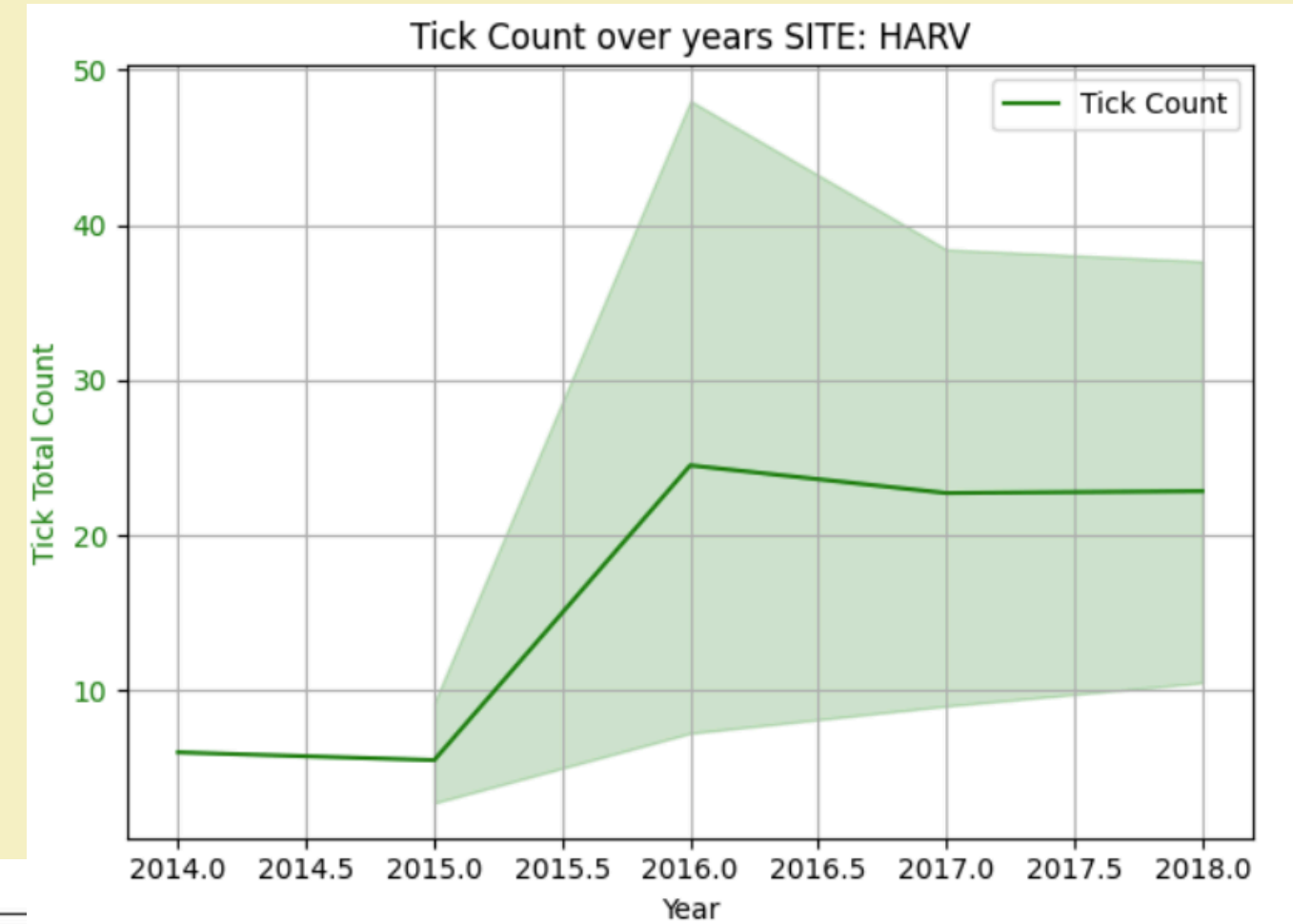
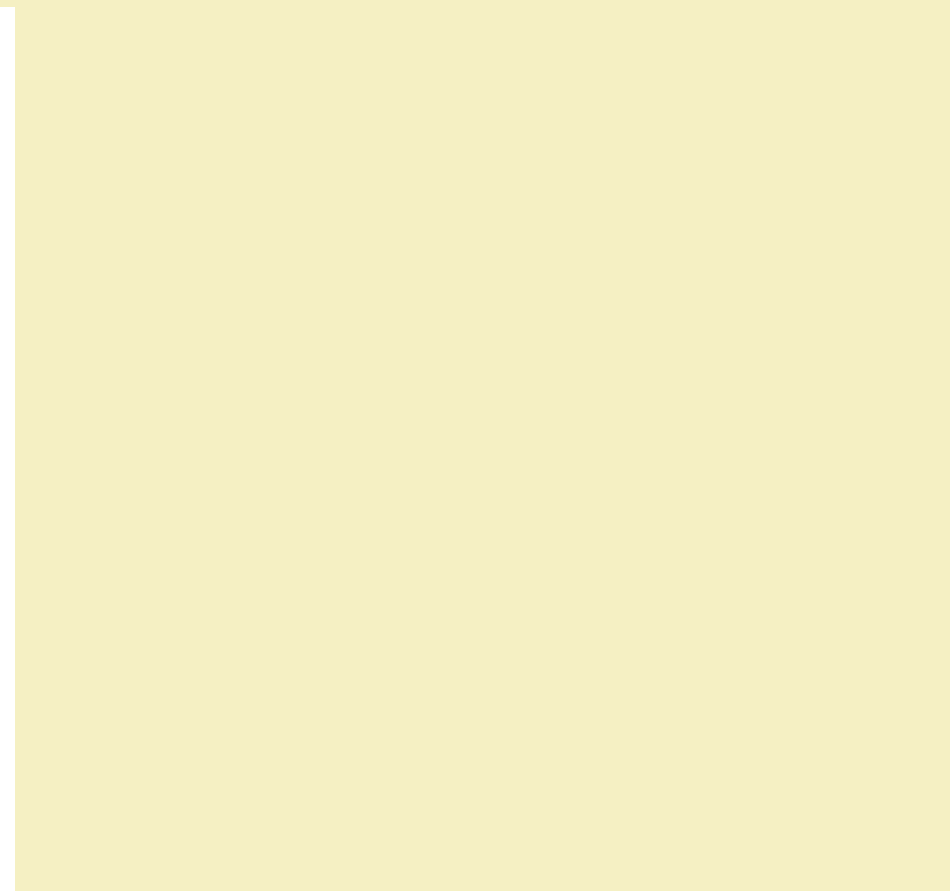
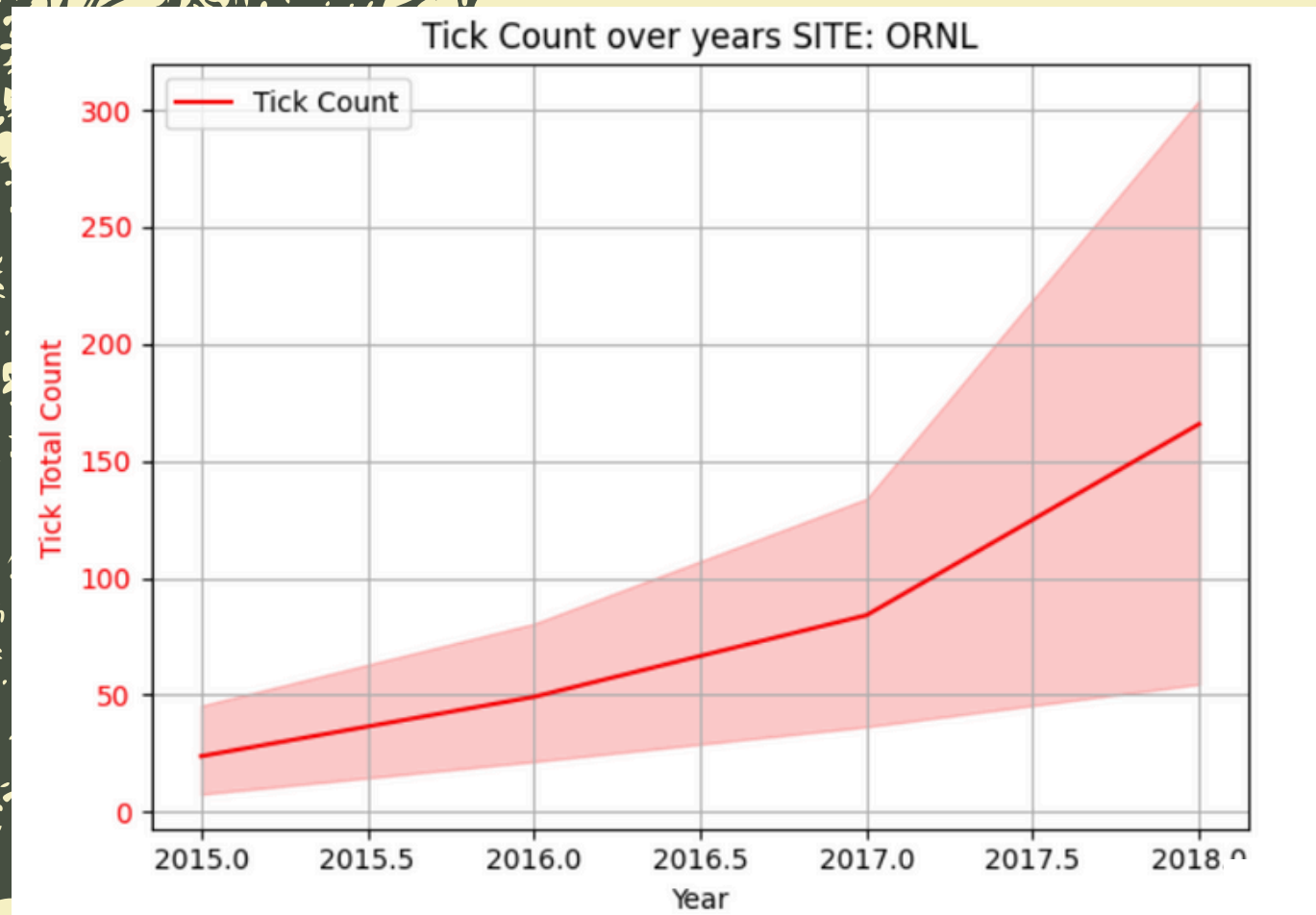
Rising temperatures, Increased humidity, and changing precipitation pattern due to climate change have contributed to the geographic expansion and population growth of ticks in the United States from 2014 to 2024, with a stronger correlation observed in the Eastern U.S. compared to the Western U.S.

METHODOLOGY



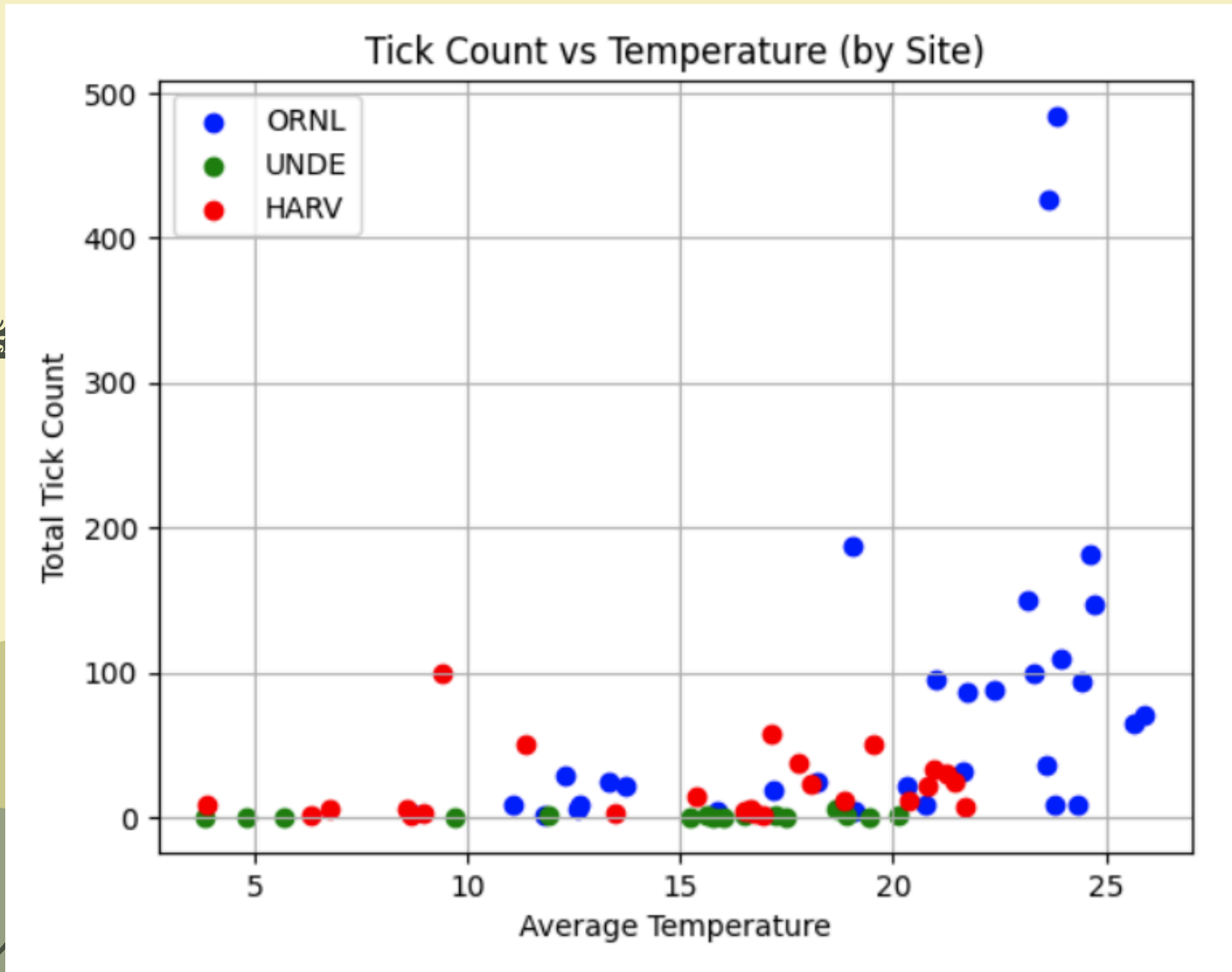
RESULTS

Change in tick population over time



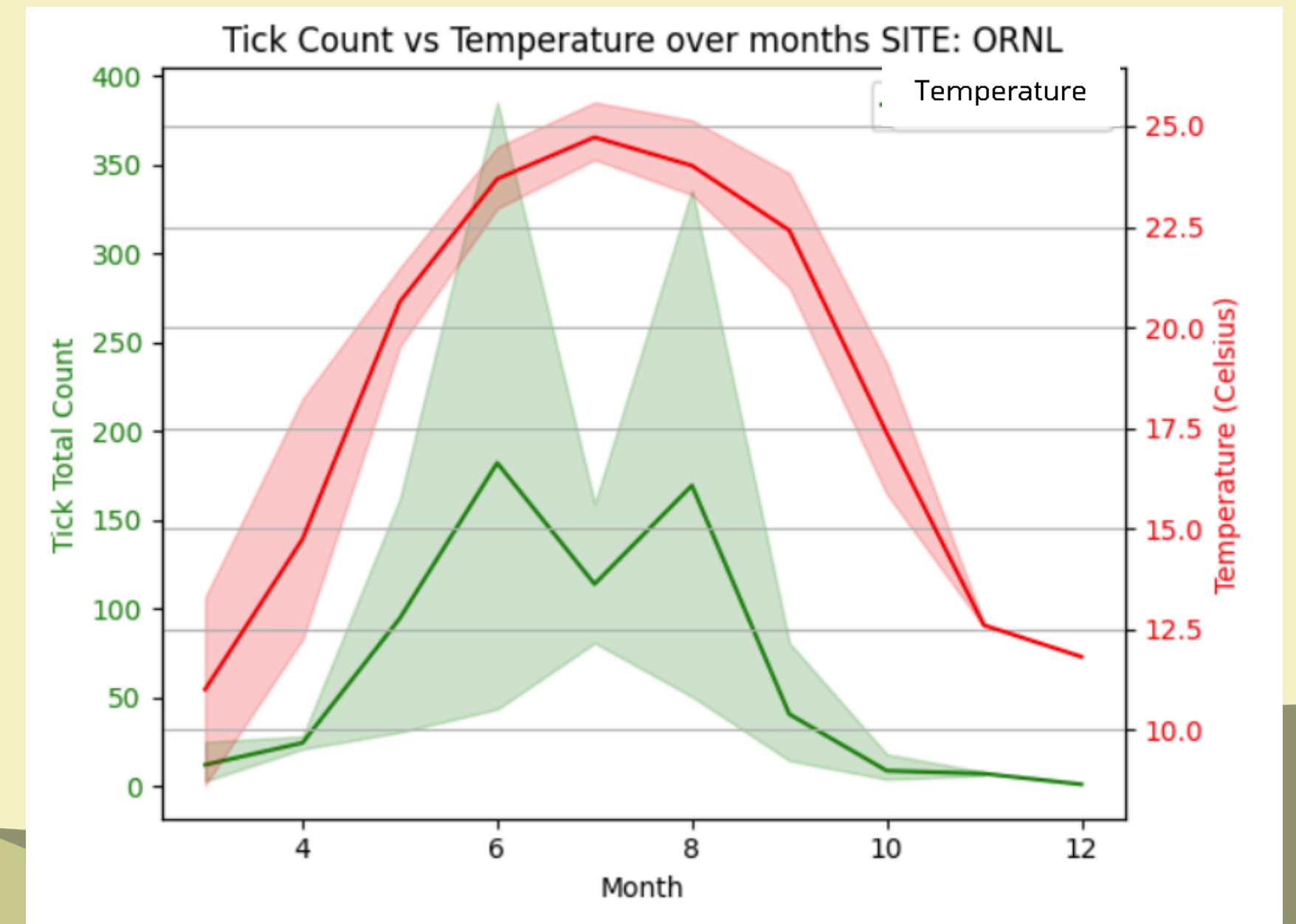
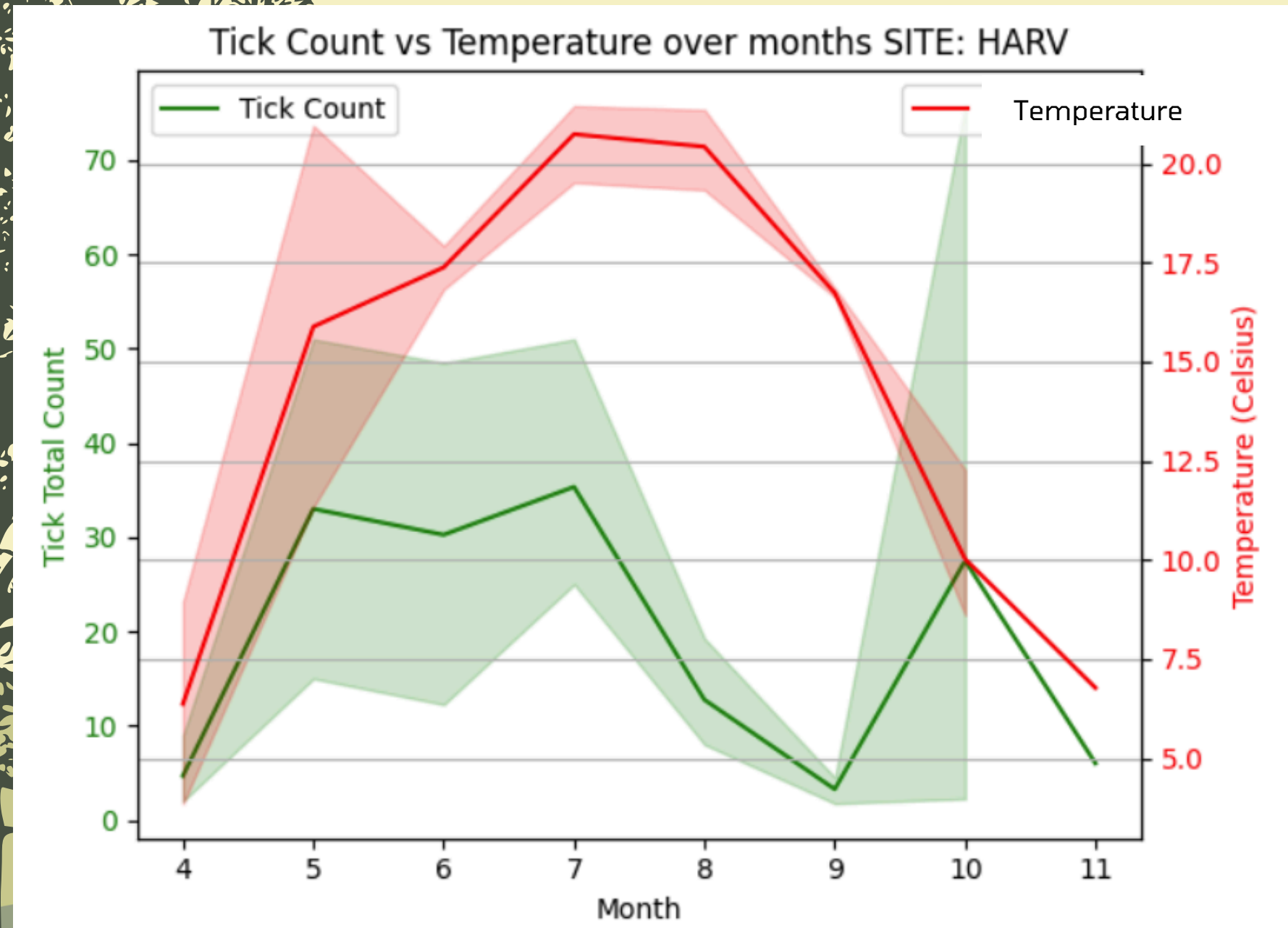
RESULTS

Tick Count vs Temperature by site



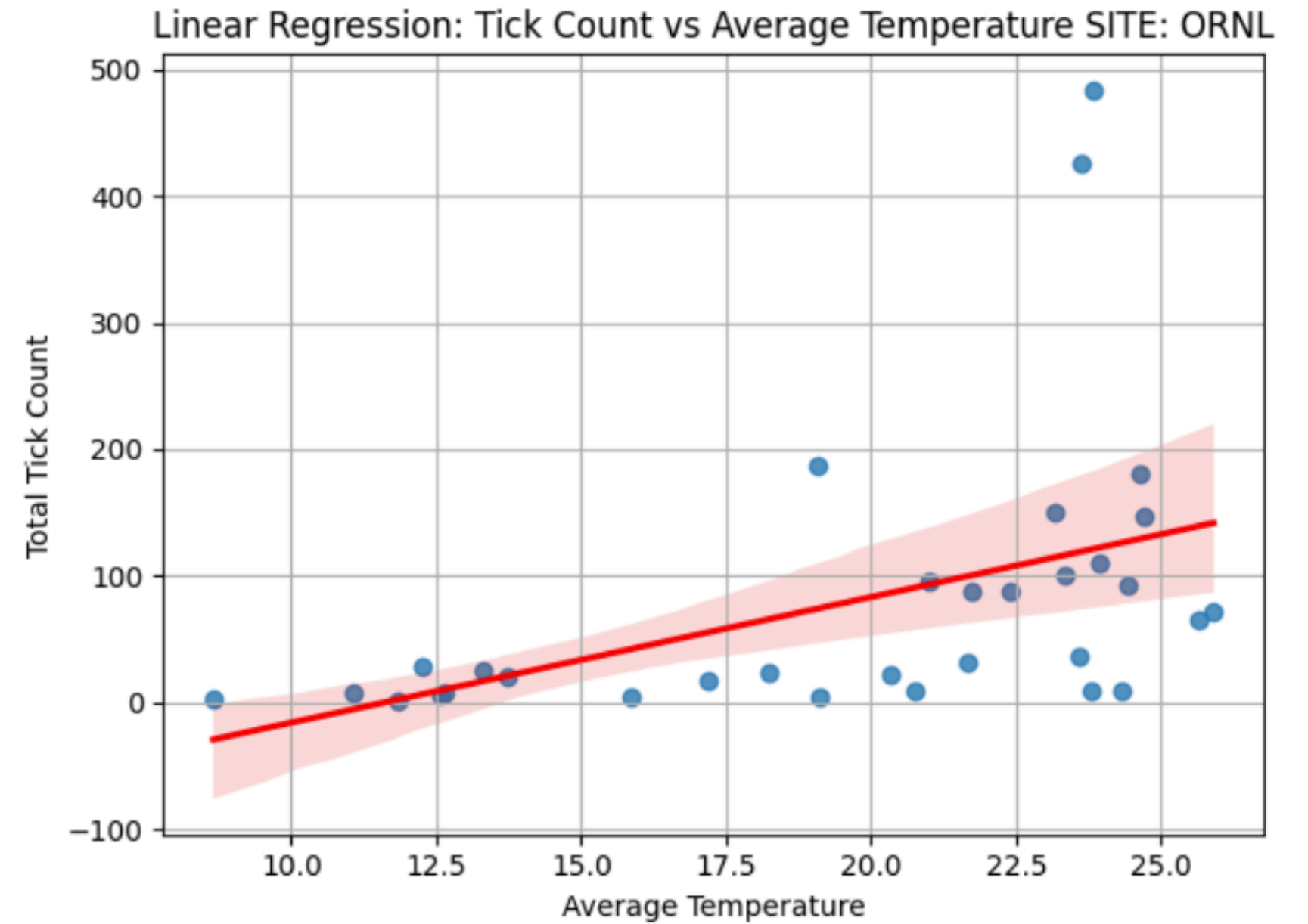
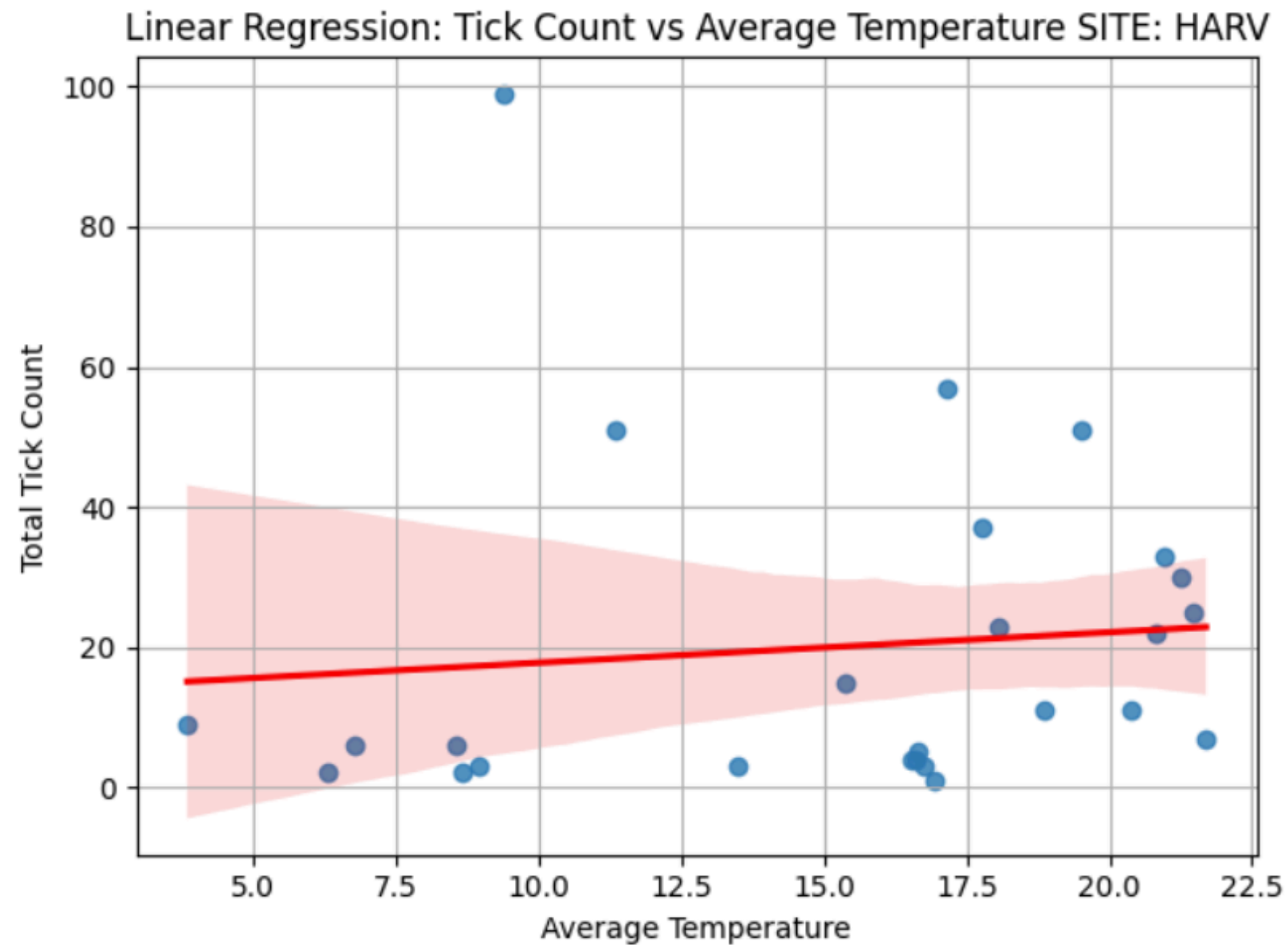
RESULTS

Tick Count vs Temperature over months



RESULTS

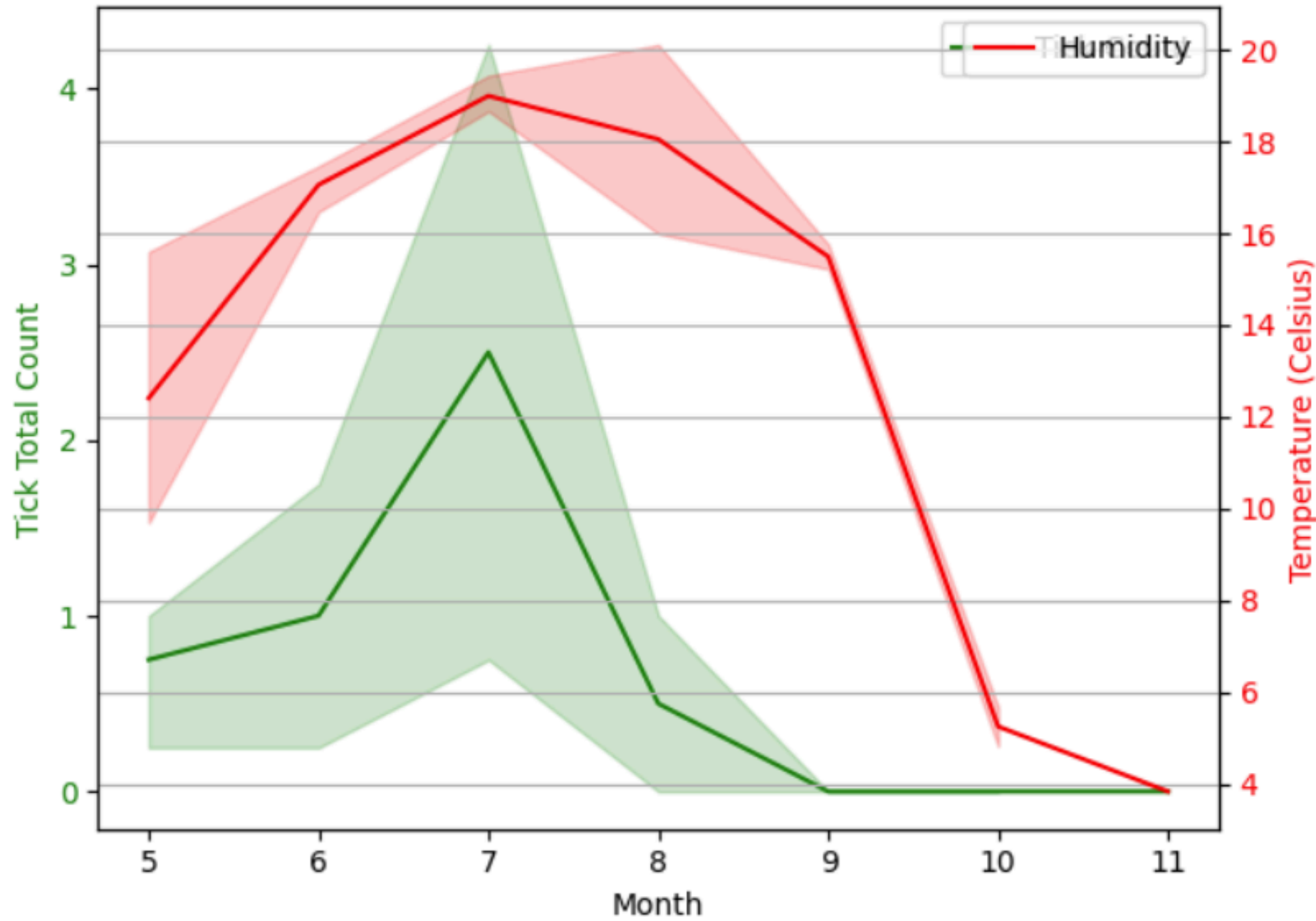
Tick Count vs Temperature over months



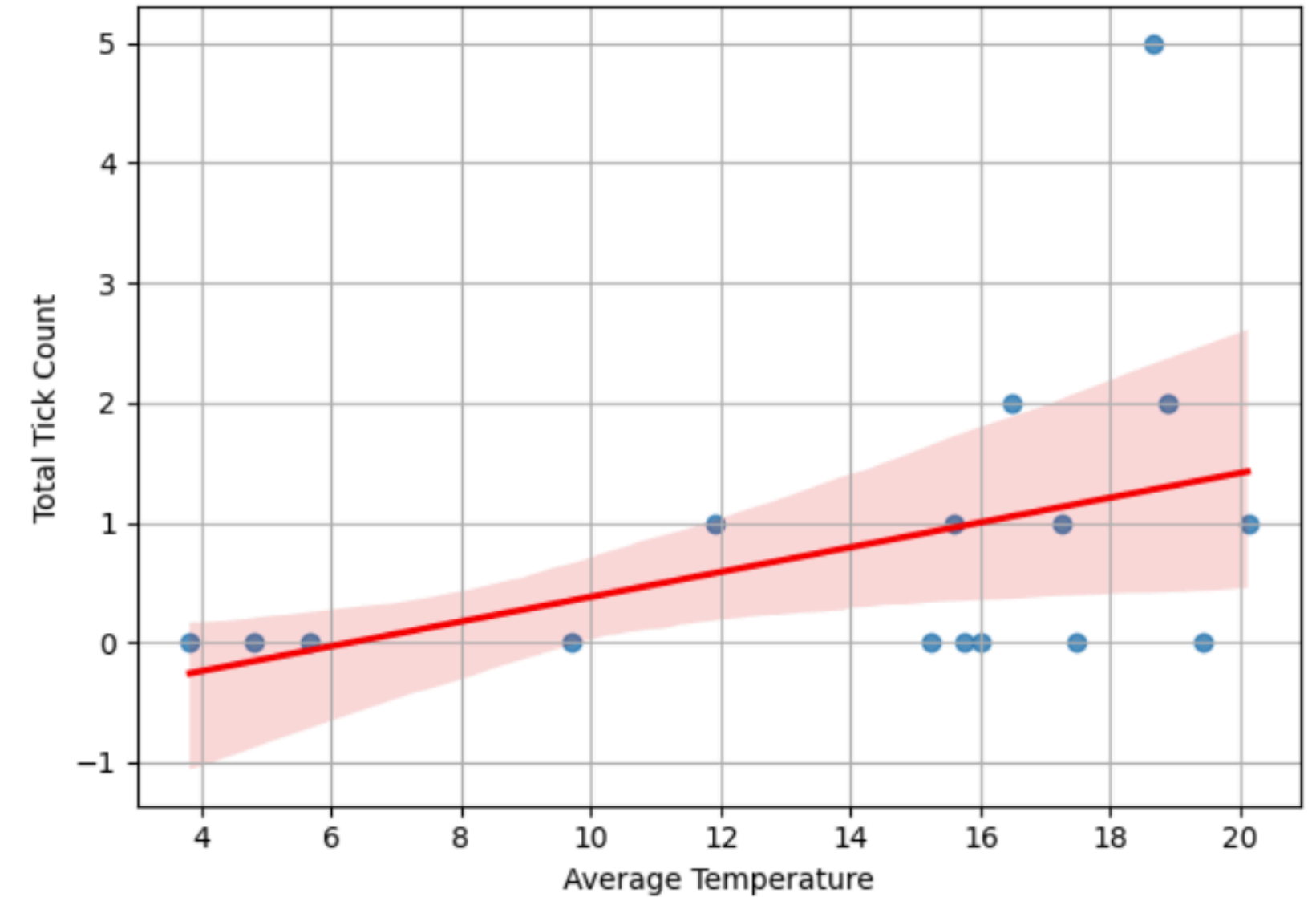
RESULTS

Tick Count vs Temperature

Tick Count vs Temperature over months SITE: UNDE

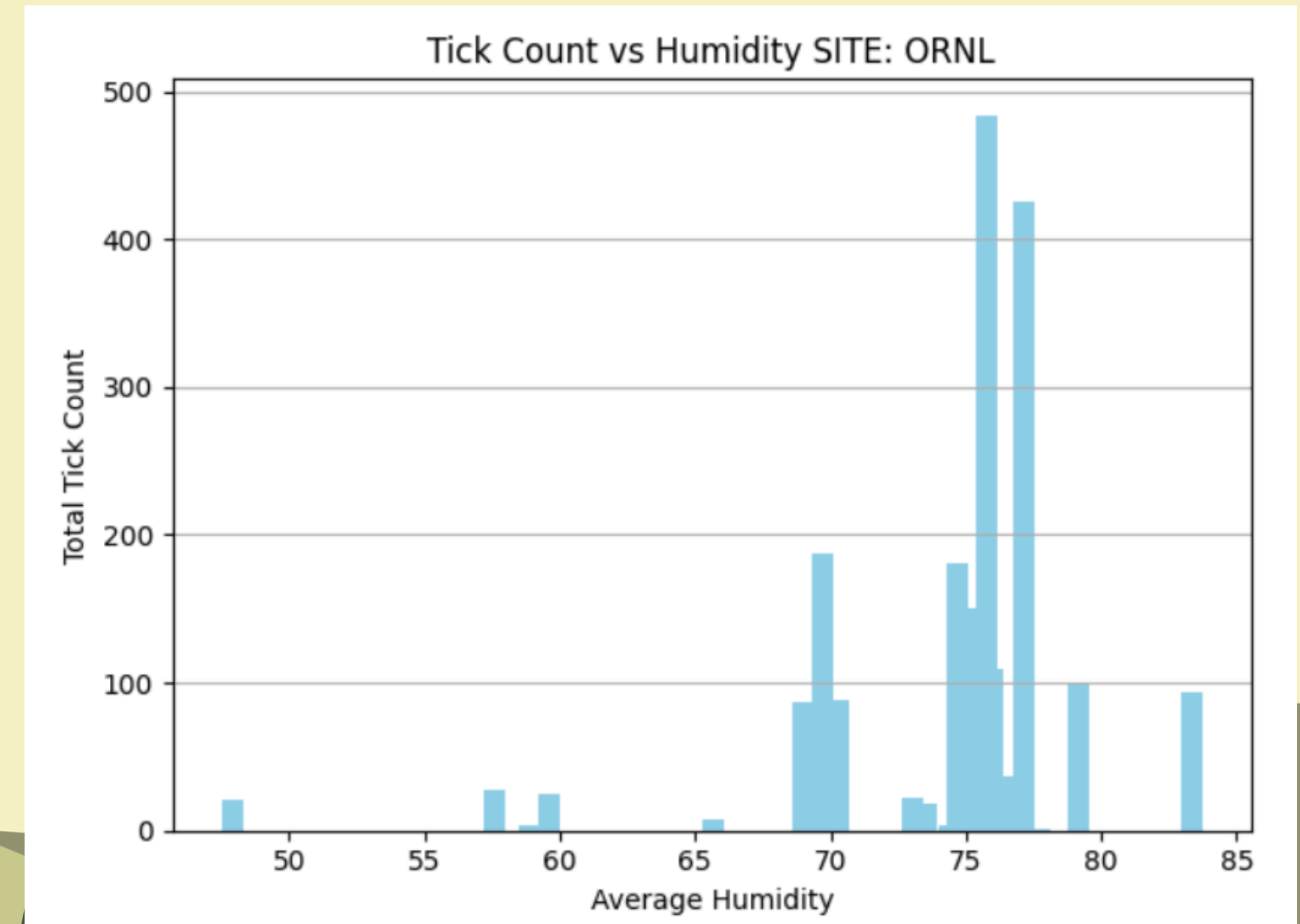
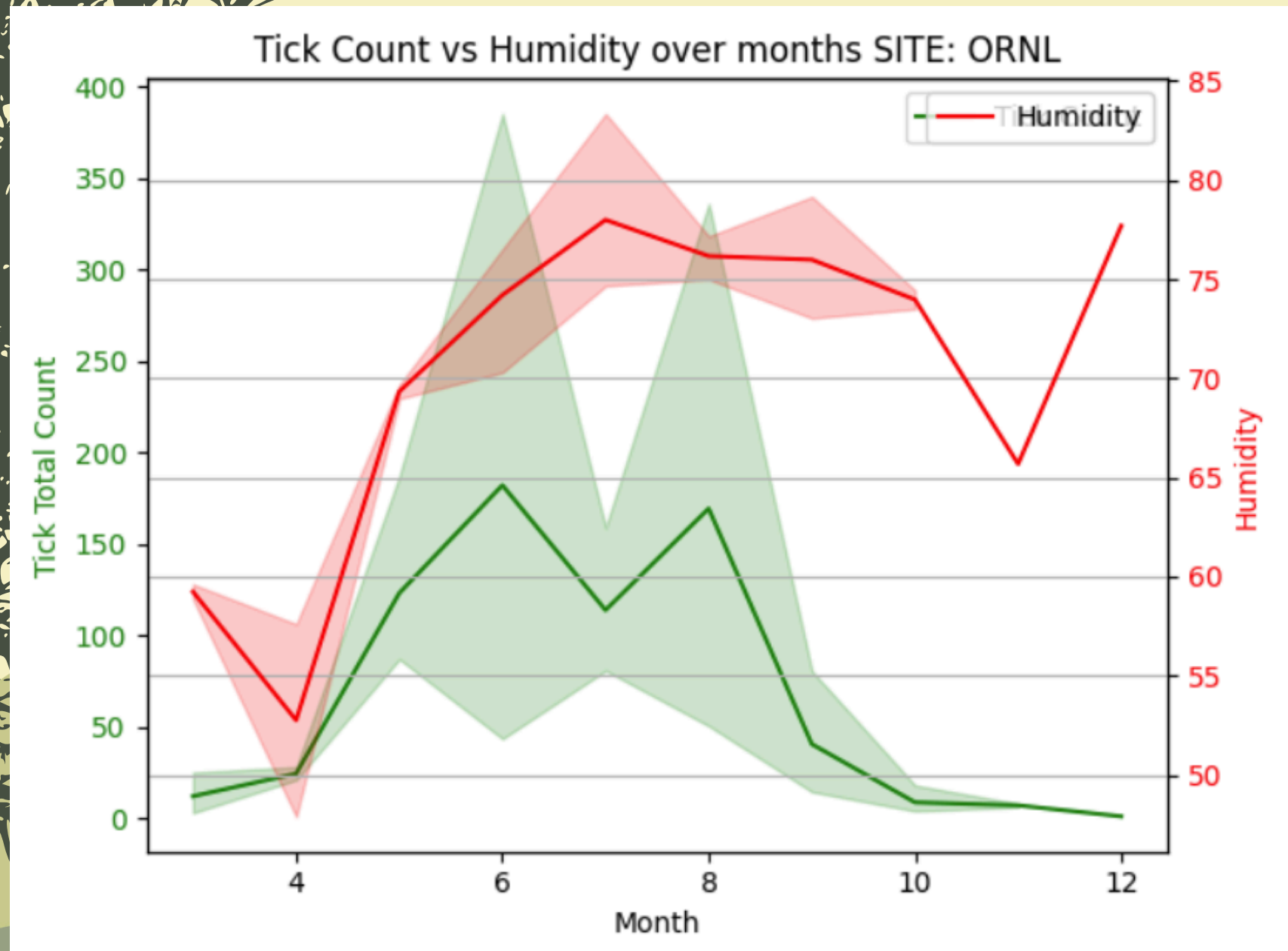


Linear Regression: Tick Count vs Average Temperature SITE: UNDE



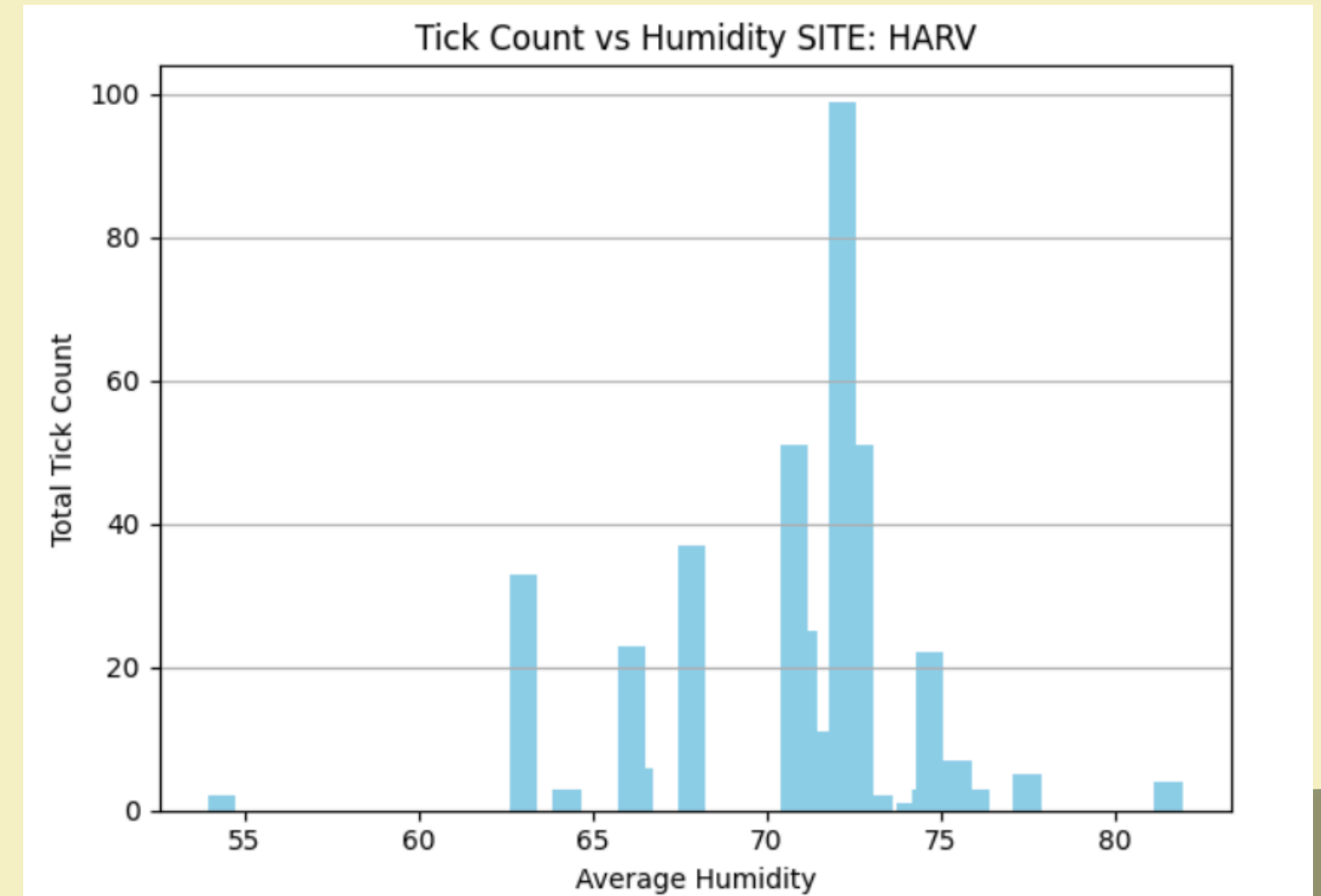
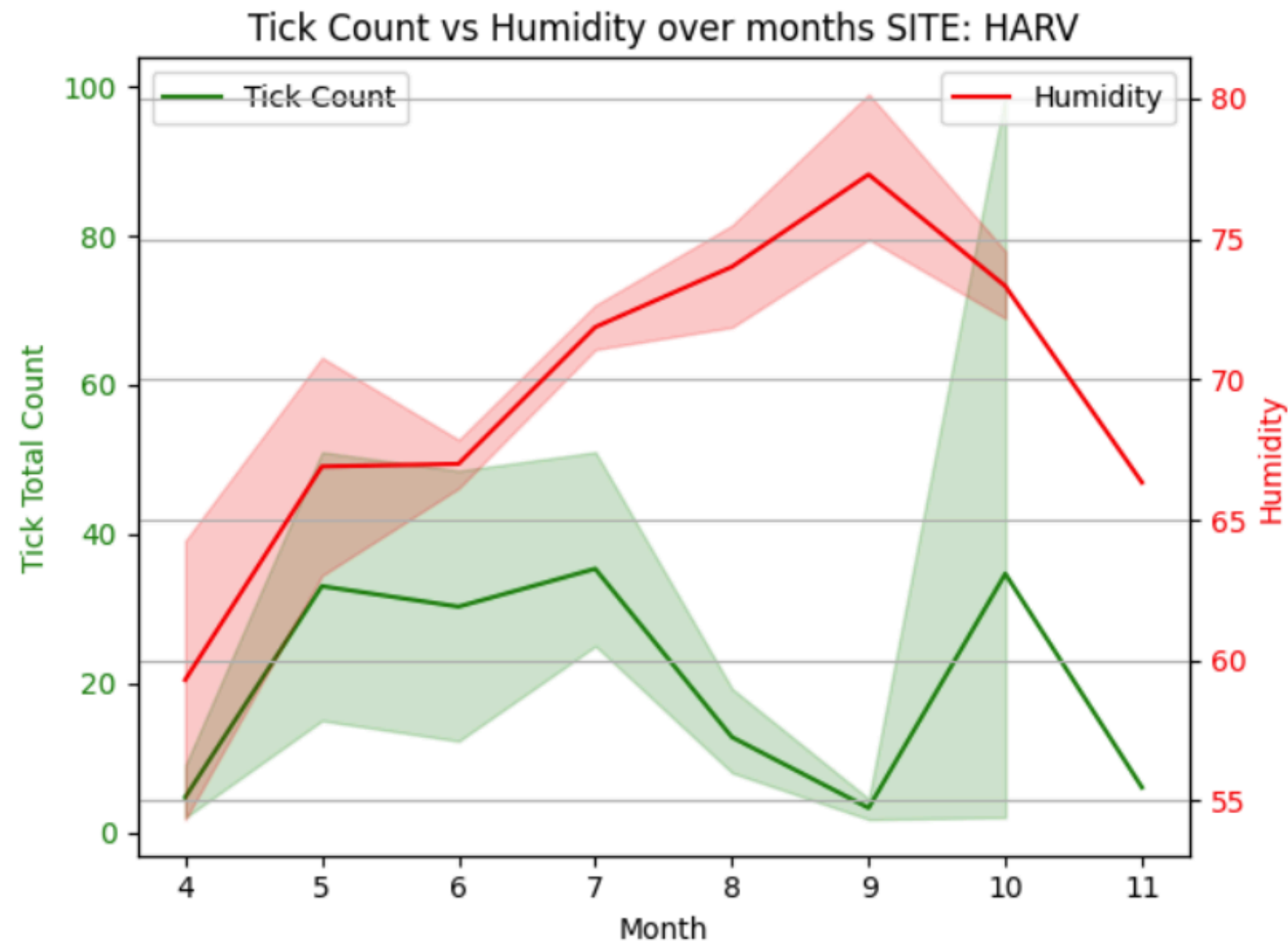
RESULTS

Tick Count vs Humidity SITE: ORNL



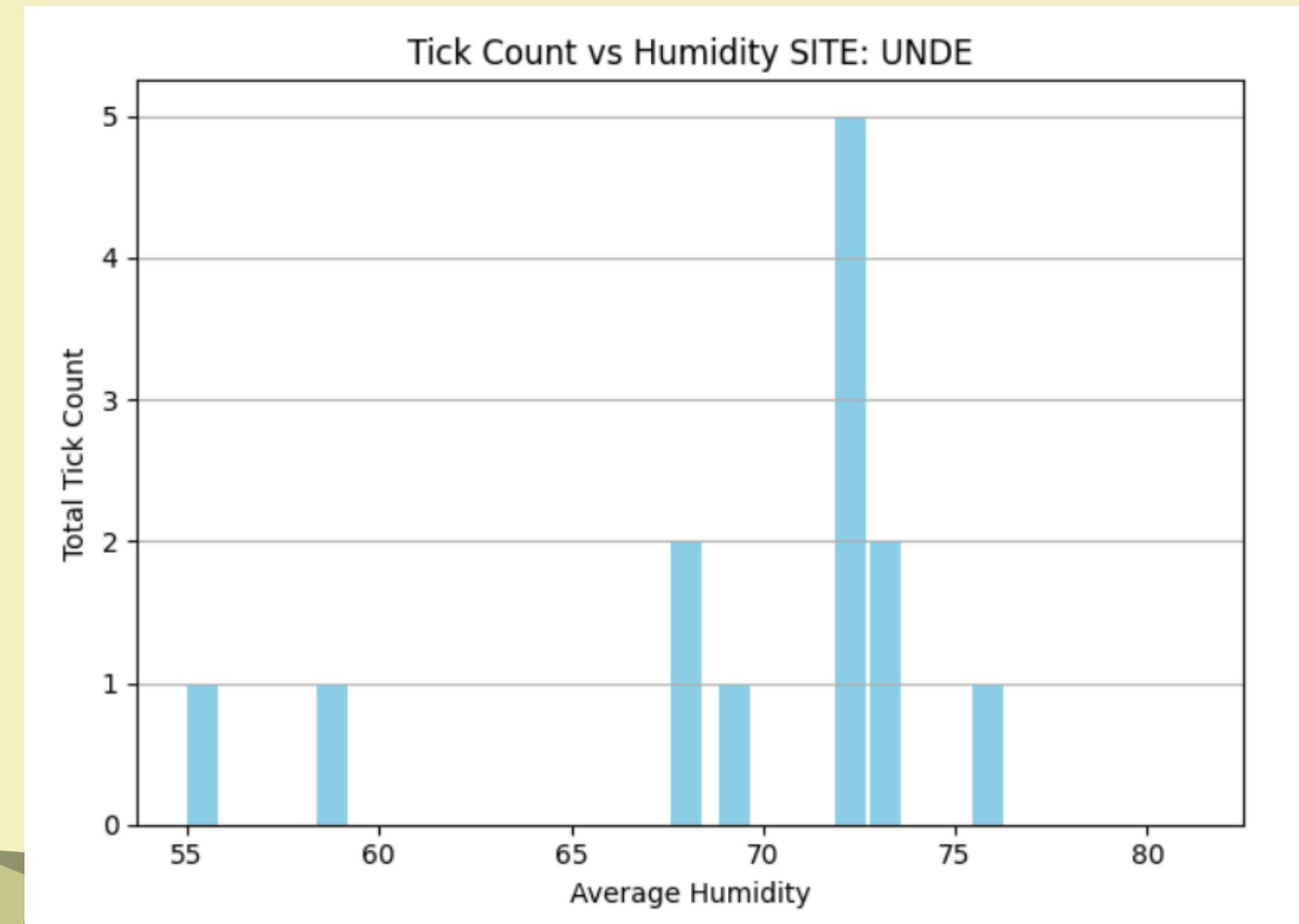
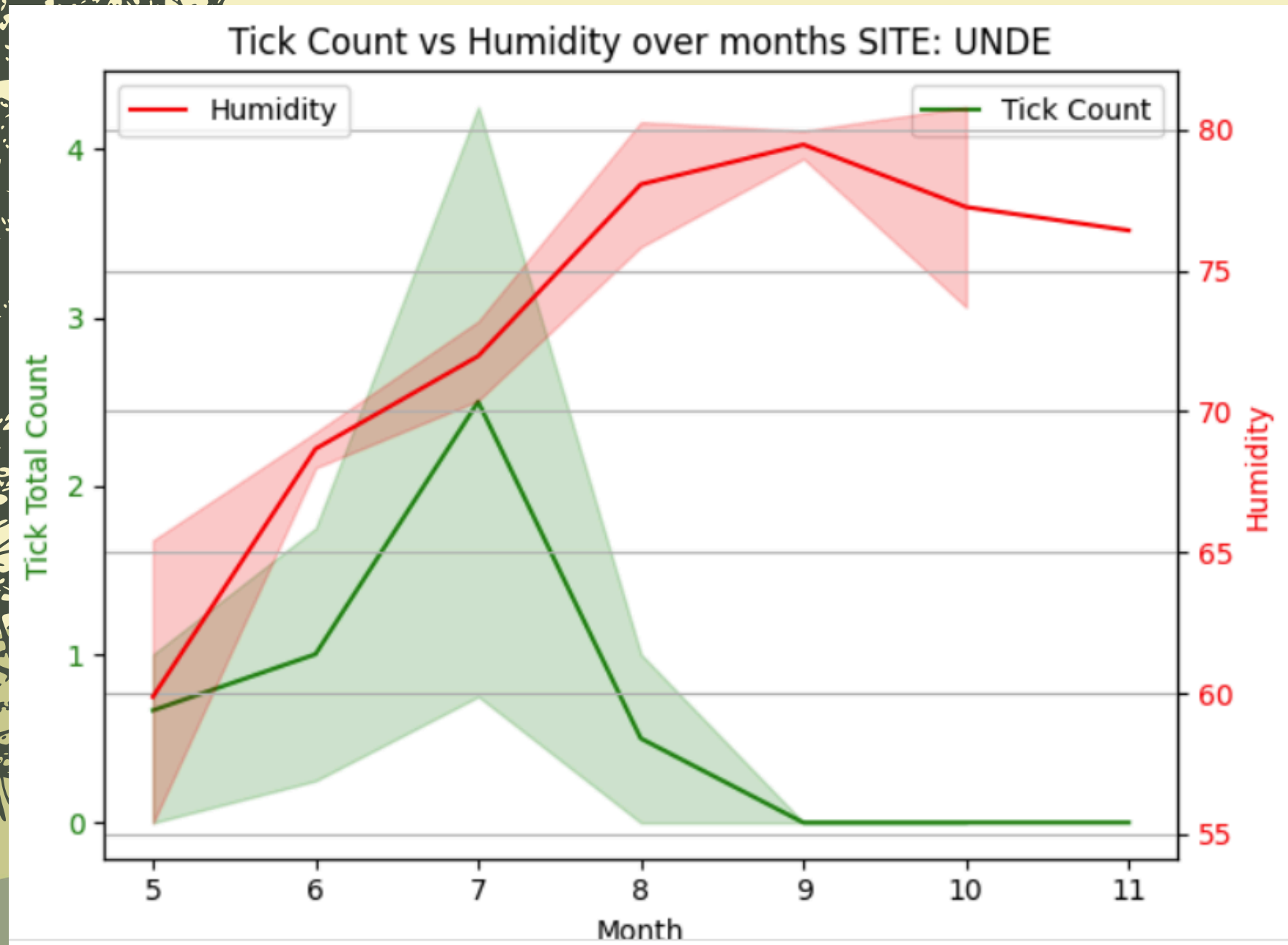
RESULTS

Tick Count vs Humidity SITE: HARV



RESULTS

Tick Count vs Humidity SITE: UNDE



DISCUSSION AND CONCLUSION

- Tick Populations show significant growth at warmer sites (ORNL)
- Optimal conditions: 20-25°C temperature, 70-80% humidity
- Observed optimal temperatures match Ogden's findings (2013)
- Humidity importance aligns with Sonenshine (2018) and Molaei (2024)



REFERENCES

- MOLAEI, G., LITTLE, E. A. H., WILLIAMS, S. C., & STAFFORD, K. C. (2024). EXPANSION OF THE GEOGRAPHIC RANGE OF Ixodes scapularis AND Amblyomma americanum IN THE NORTHEASTERN UNITED STATES. JOURNAL OF INFECTIOUS DISEASES, 226(3), 370-373. [HTTPS://DOI.ORG/10.1093/INFDIS/JIAC249](https://doi.org/10.1093/infdis/jiac249)
- OGDEN, N. H., MECHAI, S., & MARGOS, G. (2013). CHANGING GEOGRAPHIC RANGES OF TICKS AND TICK-BORNE PATHOGENS: DRIVERS, MECHANISMS AND CONSEQUENCES FOR PATHOGEN DIVERSITY. FRONTIERS IN CELLULAR AND INFECTION MICROBIOLOGY, 3, 46. [HTTPS://DOI.ORG/10.3389/FCIMB.2013.00046](https://doi.org/10.3389/fcimb.2013.00046)
- SONENSHINE, D. E. (2018). RANGE EXPANSION OF TICK DISEASE VECTORS IN NORTH AMERICA: IMPLICATIONS FOR SPREAD OF TICK-BORNE DISEASE. INTERNATIONAL JOURNAL OF ENVIRONMENTAL RESEARCH AND PUBLIC HEALTH, 15(3), 478. [HTTPS://DOI.ORG/10.3390/IJERPH15030478](https://doi.org/10.3390/ijerph15030478)